

TET4160 - Insulation Materials for High Voltage Applications

Course Project

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1 Introduction

1.1 Background and Motivation

High voltage (HV) cable terminations are essential components in the connection between underground cables and overhead lines, transformers, or gas-insulated switchgear (GIS). They ensure the electrical, mechanical, and environmental transition between different insulation systems while maintaining adequate dielectric performance and reliability.

The performance of HV terminations strongly depends on the control of electric field stress at the interface between the conductor, insulation, and surrounding gas. Improper grading of this field can lead to partial discharges (PD), electrical breakdown, or long-term degradation of the insulation. Hence, a fundamental understanding of the underlying insulation principles and discharge mechanisms is crucial.

Two main approaches are used to control electric stress in HV terminations:

1. Capacitive field grading, achieved by introducing layers of materials with different permittivities.
2. Geometrical field grading, achieved by shaping the insulation (typically with a stress relief cone) to control the field distribution.

This project focuses on the latter, i.e., a geometrical field grading termination using compressed gas as the main insulating medium. The study aims to investigate how changes in cone geometry and gas pressure influence the electric field distribution and breakdown voltage.

1.2 High-Voltage Cable Terminations

High-voltage cable terminations (commonly referred to as “cable terminations”) are the interface components that connect the end of an underground high-voltage cable to ancillary equipment such as gas-insulated switchgear (GIS), overhead lines, transformers, or other electrical apparatus. Their role is to provide mechanical support, environmental sealing, electrical insulation and stress control for the end of the cable system. [1]

The construction of a cable termination typically includes the following elements:

- The central conductor (copper or aluminium) of the cable, often with a metallic screen (tape or wire) and armour.
- The main cable insulation layer (for example XLPE or EPR).
- Semi-conductive screens (inner and outer) for the insulation, to ensure a uniform potential distribution.
- A stress relief component (often called “stress cone”) which serves to manage the end of the metallic screen and smooth the transition of equipotential lines outwards.

- The outer insulation, which may be an insulating medium (air, gas, resin) filled or dry, and associated sheds or external insulation surfaces for outdoor terminations.
- A sealing arrangement for environmental protection (moisture, contaminants, pollution) and in some cases a pressurized insulating medium.

Figure 1.1 shows an example of a cable termination with the main components mentioned above.

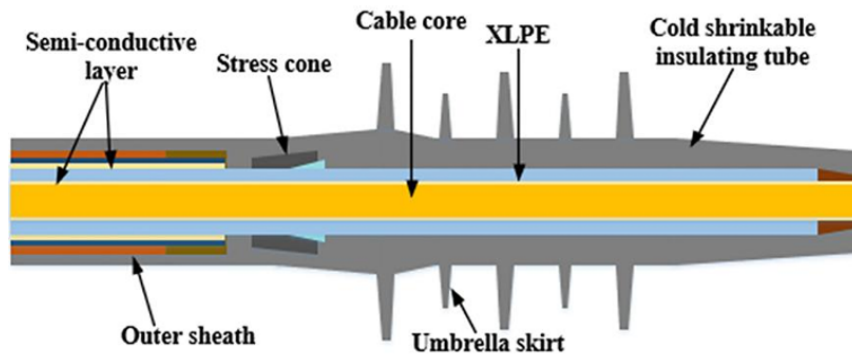


Figure 1.1: Structure of cable terminal used in distribution power systems [2]

From an electrical perspective, the key objective of the termination design is to control and minimize electric field stress concentrations, especially at the edge of the metallic screen and at the interface between insulation and external medium (air or gas). If the field is not adequately managed, local high field intensities can lead to partial discharges, surface tracking, or full breakdown.

Various insulation systems are employed at the termination (and indeed over the cable system), depending on the voltage level, environment, and design constraints. Some of the key insulation media are listed below.

- Air: in some termination designs the insulation is achieved by dry air or a pressurized air environment. This may be called “tech-air” in specific designs. The use of air (especially at increased pressure) reduces cost and environmental footprint versus traditional gases.
- Sulphur hexafluoride (SF_6): widely used in high-voltage equipment for its excellent dielectric properties and ability to withstand high field/impulse stress in compact designs. However, SF_6 has a very high global-warming potential (GWP) and regulatory environmental pressures have increased.
- g^3 (green gas for grid): this is an alternative insulating gas mixture developed to provide equivalent dielectric performance to SF_6 but with a significantly lower environmental impact (lower GWP).

The differences among these insulation media are primarily:

- Dielectric strength and behaviour under impulse and switching overvoltages.

- Pressure dependence: pressurised gases (SF_6 , g^3) often require a sealed enclosure and can achieve higher breakdown voltages at given clearances than air at atmospheric pressure.
- Environmental/operational constraints: SF_6 concerns lead to trend away from its use; g^3 and air/tech-air offer more sustainable alternatives.
- Mechanical/installation implications: use of gas may require pressure vessels, monitoring systems, whereas dry air/tech-air may allow simpler constructions but perhaps larger dimensions.

In terms of voltage levels, the design of cable terminations strongly depends on the system high-voltage rating. As the voltage class increases (for example from 132 kV to 245 kV, 400-500 kV or beyond), the required clearances, insulation dimensions, creepage distances, shape optimization and field grading become more critical. Larger voltages typically imply larger termination hardware, more extensive stress control systems, and in many cases the use of advanced insulation media (gas, composite insulators) rather than simple air-insulated terminations.

In this project, the specific scenario is a cable termination for a 500 kV system, employing tech-air (compressed air) as the insulation medium (as shown by Figure 1.2. Thus the design challenge involves ensuring that the dielectric strength of the air insulation at the termination, combined with the geometrical and material field-grading components (such as the stress relief cone), is sufficient for the 500 kV rating.

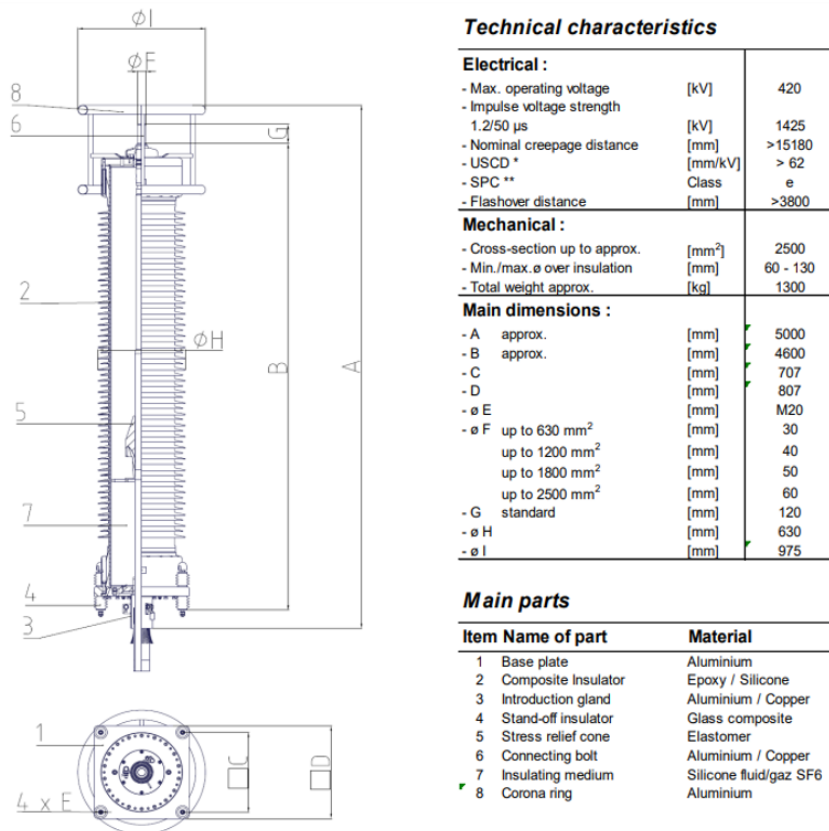


Figure 1.2: 500 kV cable termination from Nexans (FRX 1.420 - Model II 15180 - C)

1.3 Theory of Streamer Breakdown and Pressure Dependency

In the first task of the project, several theoretical aspects must be addressed: the breakdown criteria for streamers in the gas itself and along insulating surfaces; the pressure dependency of the gas breakdown; the breakdown criteria for the outside of the composite insulator (in dry air conditions); and a review of stress relief cone design (size, geometry, performance). Below follows a detailed theoretical presentation.

1.3.1 Breakdown criteria for streamers in the gas and along insulating surfaces

A streamer inception in a gaseous medium originates from the instability of an electron avalanche when the ionization process overtakes the loss mechanisms. The process is governed by Townsend's first ionization coefficient α , which depends on the electric field strength E and the mean free path of electrons λ_e . Since the mean free path is inversely proportional to the gas pressure p , the energy gained by an electron between collisions, proportional to $E \lambda_e$, scales with E/p . [1]

Therefore, the probability of ionizing collisions (and thus the ionization coefficient) can be expressed as:

$$\alpha = p \cdot f\left(\frac{E}{p}\right) \quad (1.1)$$

where $f(E/p)$ is an experimentally determined function describing the ionization behaviour of the gas. Likewise, the secondary ionization (or secondary emission) coefficient γ depends on the same reduced electric field:

$$\gamma = g\left(\frac{E}{p}\right) \quad (1.2)$$

These relations are inserted into Townsend's breakdown equation (for homogeneous fields and including secondary emission processes):

$$\gamma \left(e^{\alpha d} - 1 \right) = 1 \quad (1.3)$$

which, after substitution and algebraic manipulation, leads to the Paschen's law. Paschen's law states that the breakdown voltage U_b in a homogeneous field (for a given gas and temperature) depends only on the product $p \cdot d$:

$$g\left(\frac{U}{pd}\right) \left[e^{pd \cdot f\left(\frac{U}{pd}\right)} - 1 \right] = 1 \quad (1.4)$$

This equation describes the Paschen curve (shown in Figure 1.3), which gives the breakdown voltage as a function of $p \cdot d$.

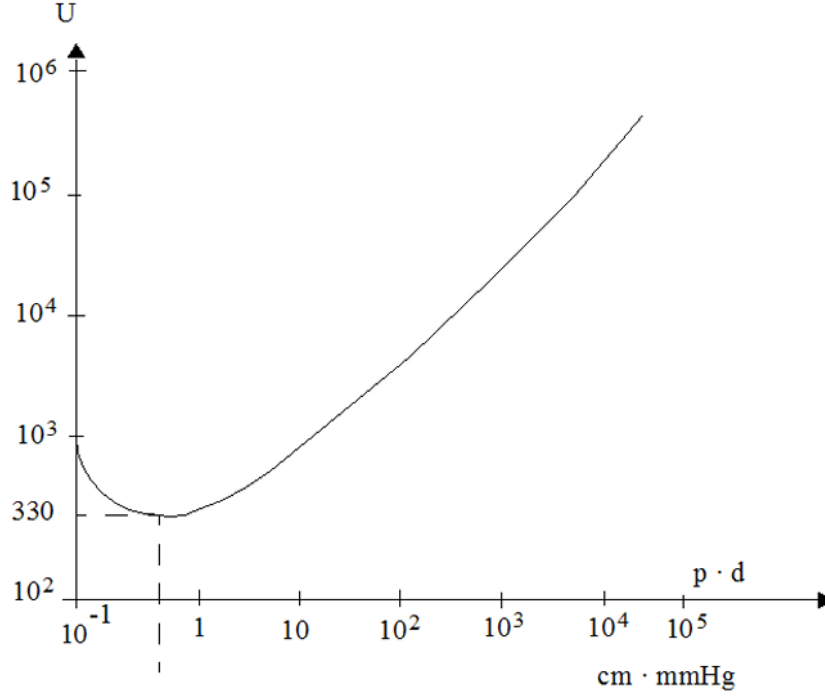


Figure 1.3: Paschen curve for air

It provides the foundation for estimating the ionization coefficient α and the effective ionization coefficient:

$$\alpha_{\text{eff}} = \alpha - \eta \quad (1.5)$$

where η is the electron-attachment coefficient (for example in electronegative gases). The value of α_{eff} determines whether electron avalanches can grow or decay under given conditions.

Once α_{eff} becomes sufficiently large for the avalanche to contain about 10^8 electrons, space-charge effects distort the local field and initiate a self-propagating streamer.

The breakdown criterion for streamers in the gas itself is then often expressed as:

$$\int_0^d \alpha_{\text{eff}}(E, p) dx = K \quad (1.6)$$

with $K \approx 10-20$ for air at atmospheric pressure (this value is empirical, and may vary for different gases and conditions). [3]

When a streamer develops along an insulating surface, the discharge is guided by the gas–solid interface rather than purely through the bulk gas. In this case, the tangential electric field E_t dominates the ionization process along the surface path, while the normal field E_n contributes to charge accumulation and surface charging. The presence of the surface modifies the local electric field through dielectric polarization and the accumulation of positive and negative charges, thereby creating conditions favourable

for surface streamer inception.

The growth of the streamers along the surface can still be described by an effective ionization coefficient defined as:

$$\alpha_{\text{eff, surf}} = \alpha_t(E_t, p) - \eta_t(E_t, p) \quad (1.7)$$

where

- $\alpha_t(E_t, p)$ is the ionization coefficient due to the tangential component of the field along the surface;
- $\eta_t(E_t, p)$ is the corresponding attachment coefficient (or loss coefficient) for the surface-path ionization.

Both coefficients depend on the local tangential field E_t , which is often enhanced near the surface due to field distortion, surface roughness, charge deposition or triple-point effects.

The breakdown (surface streamer) criterion is then formulated analogously to the gas-phase case:

$$\int_0^L \alpha_{\text{eff, surf}}(E_t(x), p) dx = K_s \quad (1.8)$$

where

- L is the discharge path length along the surface;
- K_s is an empirical constant depending on the surface material, humidity and contamination level.

Physically, this condition means that when the net ionization integrated along the surface exceeds a critical threshold K_s , the avalanche becomes unstable and transforms into a self-propagating surface streamer. These streamers are sustained not only by ionization in the gas but also by secondary processes such as photo-emission, surface-charge release and local field enhancement at the interface.

In the context of our project (a 500 kV cable termination employing tech-air as insulating medium) these criteria are particularly relevant because:

- The breakdown in the bulk gas region between the termination cone and outer enclosure must be avoided; thus the integrated ionization criterion in the gas gives a threshold for safe field levels at a given pressure.
- The outer surface of the composite insulator (in air) must be evaluated with respect to surface-streamer inception; thus the tangential surface field must be limited in design to maintain $\int \alpha_{\text{eff, surf}} dx < K_s$.
- Variations in gas pressure (changing p) directly affect the mean free path and thus the scaling of $\alpha \propto p f(E/p)$. Hence modelling in the next sections will vary pressure and evaluate how close the

field-integral comes to these thresholds.

By combining these theoretical foundations with the geometry of the termination and the dielectric materials chosen, we can set criteria for model design: the maximum local electric field (and its distribution) should be sufficiently below the streamer-inception boundary in both the bulk gas and along the outer surface.

1.3.2 Gas pressure dependency

The dependency of gas pressure on breakdown behaviour is based in the same physical foundations previously discussed (ionization, avalanche growth, streamer inception), but here we focus specifically on how varying the gas pressure p influences the ionization coefficient, mean free path of electrons and, consequently, the breakdown voltage of a gaseous gap.

As said before, the Townsend ionization coefficient α in gases can be expressed in terms of the reduced electric field E/p , as written in Equation 1.1.

These relations imply:

- When the pressure p increases, the mean free path λ_e becomes shorter, so electrons gain less energy between collisions, which tends to reduce ionization (for a given E).
- When the pressure p decreases, λ_e increases, electrons can gain more energy between collisions, making ionization easier.
- Consequently, the breakdown voltage U_b in a homogenous gap becomes a function of the product $p \cdot d$ (where d is the gap length), which is the content of Paschen's law.

This means that for a given geometry the breakdown condition depends not individually on p or d alone, but on their product $p \cdot d$. At moderate to high pressures and gaps, one often finds an approximately linear scaling of breakdown field with pressure (i.e., $E_{bd} \sim p$ for fixed gap) though departures occur in the low-pressure or very small gap regimes.

In summary, the gas pressure dependency directly influences the ionization rate (via α), the avalanche growth, and thus the breakdown voltage in gases. In our context of a gas-insulated termination (tech-air) for a 500 kV cable termination, this implies that increasing the internal pressure of the tech-air medium will elevate the breakdown threshold (for a given geometry), thereby allowing smaller clearances or lower local fields, and reducing the margin against streamer inception.

In the modelling phase we will exploit this relationship by varying the pressure p of the insulating gas domain, and investigating how the maximum local electric field in the termination relates to the reduced field E/p and the corresponding proximity to the streamer inception criterion introduced previously.

1.3.3 Breakdown criteria for the outside of the composite insulator in dry air

The breakdown criteria for air around a composite insulator, under non-uniform electric-field conditions like those present in a high-voltage termination, is primarily based on the streamer criterion (streamer-breakdown criterion). This criterion, unlike the simpler Townsend criterion (applicable for uniform fields and bulk gas gaps), is more appropriate for non-uniform fields and long discharge distances along the insulator surface.

- **Streamer Criterion:** breakdown does not occur simply when the electric field reaches a fixed value, but when the integration of the net ionisation coefficient (the difference between ionisation by collision and electron attachment) along a sufficiently long path reaches a critical value. For a uniform field in clean dry air at atmospheric pressure, the breakdown electric field is approximately 3 kV/mm (peak value) for bulk air. However, near an insulator's surface, this value can be lower due to surface imperfections, roughness, and the presence of pollution or charge accumulation, which facilitate the formation and propagation of streamers.
- **Surface Effects:** the insulator's surface significantly influences the breakdown process. The presence of surface charges, dielectric polarisation, and the non-uniformity of the field due to the insulator's geometry (triple junction between conductor-insulation-air) can significantly reduce the flashover voltage (surface discharge). Surface flashovers therefore usually occur at a lower tangential electric field compared to the breakdown of free air gaps. The main aspects that strongly affect the flashover path and lower the required voltage for breakdown are surface charge, roughness, contamination and humidity.

In practical terms, for a composite insulator in dry clean air conditions, one may express the surface-streamer criterion as:

$$\int_0^L \alpha_{\text{eff, surf}}(E_t(x), p_{\text{air}}) dx = K_s \quad (1.9)$$

where:

- L is the discharge path length along the surface (creepage/leakage path),
- $\alpha_{\text{eff, surf}}(E_t, p)$ is the effective ionisation coefficient along the surface (tangential field component),
- K_s is an empirical constant depending on surface material, cleanliness, humidity, contamination etc.

Because the field is non-uniform, the local tangential field $E_t(x)$ may vary significantly, and the effective path length may be influenced by shed geometry and surface profile.

In the context of our project the following considerations apply:

- The outer surface of the termination must ensure that the tangential electric field remains low enough

so that the integrated ionization along the surface path does not reach the critical threshold for surface streamer inception.

- Since surface conditions (contamination, humidity, ageing) can degrade performance, the design must include safety margins relative to the nominal 3 kV/mm value for bulk air, recognising that local surface-flashover values may be lower.
- The geometry of the termination outer surface (creepage distance, sheds, profiles) and the outer insulation medium (air) influence the field distribution and hence the value of $E_t(x)$ and the path length L . A proper shape and grading are essential to maintain acceptable surface stress levels.

In summary: for the outside of the termination in dry air, the breakdown criterion is essentially that the local tangential field and path length must be such that the integrated ionization remains below the threshold and that field peaks due to surface imperfections, charge deposition or pollution are avoided. This ensures reliable long-term performance of the termination surface under high-voltage stress.

1.3.4 Review of stress relief cones

The stress relief cone is an essential component in high-voltage cable terminations that use geometric field grading. Its main function is to redistribute and uniformize the electric field lines near the cut of the cable's shielding, where a dangerous concentration of the electric field would otherwise occur.

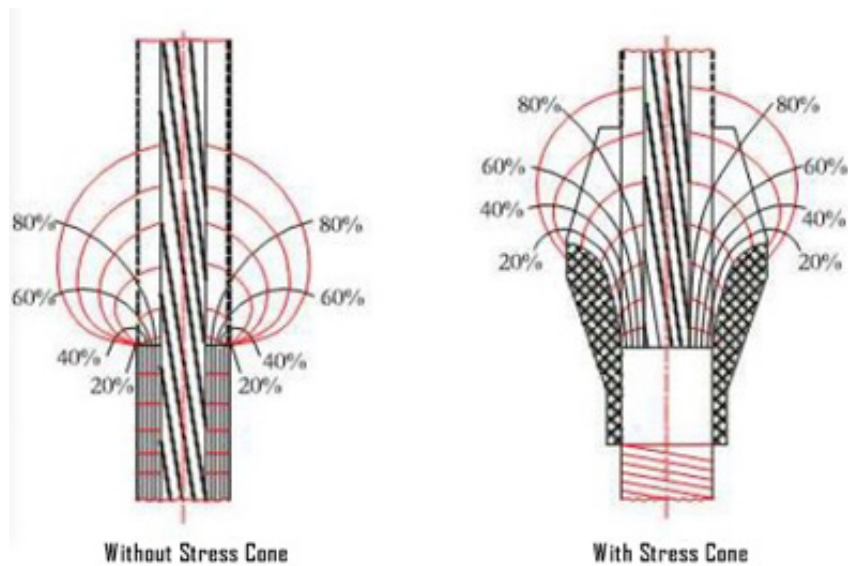


Figure 1.4: Cable termination without and with the stress cone [4]

- Principle of operation: the shielding of a high-voltage cable — which is normally kept at ground potential — ends abruptly at the termination point. Without any grading device, this abrupt end results in a region of very high electrical stress at the interface between the semi-conductive screen and the surrounding insulation or external medium. The stress relief cone, typically made of an insulating material and shaped in a bell (or conical profile), is placed around the termination of the semi-conductive

screen and insulating layer of the cable. By gradually increasing the radial (and often axial) distance between the high-voltage conductor region and the grounded screen termination, and by smoothly guiding the equipotential lines away into the bulk insulation or external medium, the cone redistributes the field over a larger volume. This design prevents local electric-field peaks that might otherwise trigger partial discharges or dielectric breakdown.

- Material and design: modern stress relief cones are commonly made of elastomeric materials such as Ethylene Propylene Diene Monomer (EPDM) or silicone rubber. These materials combine high dielectric strength, mechanical flexibility (important for installation and thermal cycles) and good resistance to environmental ageing.

In the context of our project, the design and modelling of the stress relief cone is central to the electric-field optimization. In practice:

- We need to define an initial cone geometry (based on the drawing provided in the project description) with approximate dimensions and select likely material properties (e.g., permittivity) consistent with EPDM or silicone.
- In the modelling phase we will vary the shape and the main dimensions of the stress cone and assess how these variations affect the maximum electric field in the adjacent gas/air insulation volume and at the outer termination surface.
- The objective is to ensure that the maximum field around the screen termination remains below the streamer-inception threshold in the gas and below the surface-breakdown criteria on the outer insulation, thereby ensuring reliable performance.

This review of stress relief cones thus sets the theoretical and practical foundation for the subsequent section on model construction and analysis, where the cone geometry optimization will be explicitly evaluated.

1.4 Importance of the Study

The study of a 500 kV cable termination with tech-air insulation is of great importance for several reasons:

- As transmission systems push to higher voltages (500 kV and beyond) and more compact/efficient cable terminations are required, the ability to reliably grade and manage electric fields becomes critical for reliability and longevity of grid infrastructure.
- Using tech-air (compressed air) as insulation medium provides a potentially more environmentally friendly and simpler alternative to SF₆ or other high-GWP gases, reducing environmental impact and maintenance complexity.
- Understanding how geometrical field-grading elements (such as the stress relief cone) and operational parameters (such as gas pressure) influence electric field distribution allows optimization of termination

design, potentially leading to cost savings (smaller clearances, less insulation material) and improved performance.

- For operators and designers, the ability to assess the variation of termination performance with respect to gas pressure, cone geometry, and external surfaces is essential to guarantee that the termination will perform under rated conditions, overvoltages, environmental stress and over its lifetime.

2 Voltage Levels Applied to the Cable Termination

In high-voltage cable terminations, such as the one considered in this project (500 kV system), it is crucial to understand the different voltage levels the termination must withstand. These levels include:

- **Rated system voltage (U_m):** the nominal operating voltage of the cable system (in our case 500 kV) which determines the basic insulation requirements and clearances.
- **Lightning Impulse (LI) voltage level:** a very short voltage surge (for example 1.2/50 μ s) that the termination must withstand under the standard requirements (e.g., as specified by IEC 60071-1).
- **Switching Impulse (SI) voltage level:** shorter, higher amplitude surges (e.g., 250/2500 μ s waveform) due to switching operations, to which the termination must also be designed.
- **Temporary Over-Voltages (TOV):** voltage levels slightly above nominal occurring for short durations; though lower in amplitude than impulses, they set important design margins.

2.1 Voltage Levels Specific to a 500 kV Tech-Air Termination

Because our project is based on a 500 kV cable termination using tech-air insulation, the relevant voltage levels to consider are listed below (based on IEC 60071-1) [5]. The values given in Table 2.1 are taken from the standard and refer to a highest voltage for equipment U_m of 550 kV, even though the operating voltage for our project is 500 kV.

Parameter	Value [kV]
Operating voltage U_m	500
Lightning impulse voltage (LI, 1.2/50 μ s)	1425
Switching impulse voltage (SI, 250/2500 μ s)	1175
Temporary over-voltage (TOV)	630

Table 2.1: Representative voltage levels for a 500 kV cable termination, reproduced from [5], [6]

2.2 Implications for Model Construction and Design

The declared voltage levels impose that the termination geometry, insulation medium (tech-air) and field-grading components (including the stress-relief cone) must be dimensioned and modelled not only for the nominal operating voltage but for the most severe conditions (impulse and over-voltage).

With these voltage levels defined and understood, the following modelling and analysis section can apply them to the geometry, materials and electric-field simulations for the termination.

3 Model Construction and Analysis

3.1 Model Construction in COMSOL

The model was constructed using COMSOL Multiphysics in axisymmetric (2D) configuration to represent the terminal section of the 500 kV cable termination employing tech-air insulation and an EPDM stress-relief cone. In order to realize the model in a simplified yet representative manner, some modifications and guidelines were adopted including:

- the glass-fibre wall was modelled with a thickness of 10 mm;
- the sheds (skirts) of the SiR housing were not drawn explicitly (instead the housing was replaced with a 5 mm thick SiR layer outside the glass-fibre section);
- the bottom plate was assumed circular so that rotational symmetry could be applied;
- the cable insulation was assumed to extend all the way up to the top lid, since the axial coning/diameter change seen in the older drawing is no longer current;
- the outer semiconductor termination end was placed at approximately 1000 mm above the bottom plate (consistent with typical termination arrangements).

These simplifications allowed a computationally efficient model while preserving fidelity to the geometric and field-grading behaviour expected in the real termination. The figure 3.1 shows an overview of the entire model built in COMSOL.

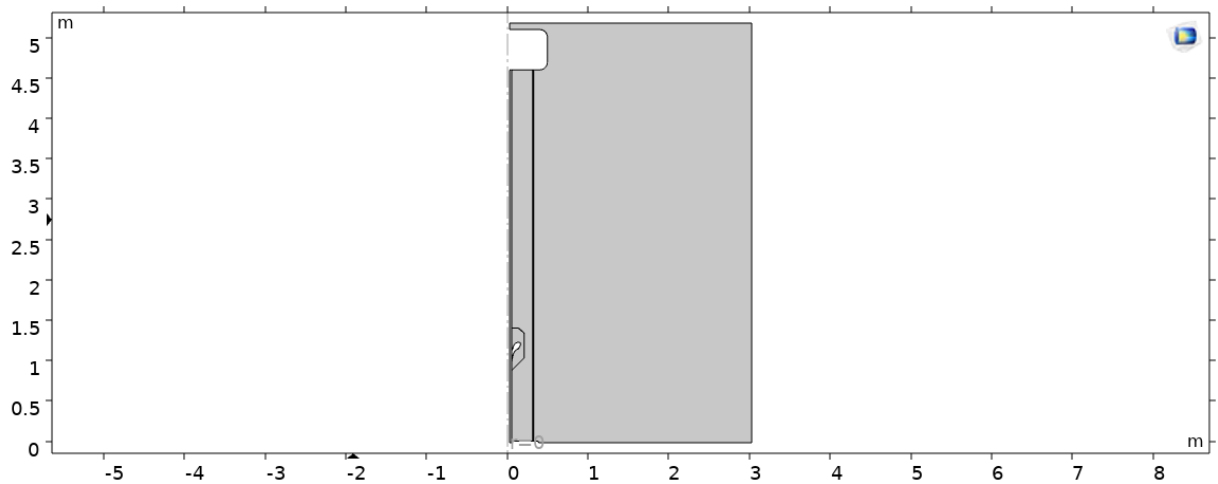


Figure 3.1: Model of the cable termination built in COMSOL

3.1.1 Geometry

The geometry includes:

- The central conductor (solid aluminium or copper) terminated at the interface.
- The cable main insulation (e.g., XLPE) around the conductor.
- The semiconductive screen termination end.
- The stress-relief cone (made of EPDM) of specified shape (length L , opening angle α , tip radius r).
- The surrounding tech-air gas region (insulating medium) up to outer grounded shield or enclosure.
- The outer composite insulator surface in dry air for the external medium.

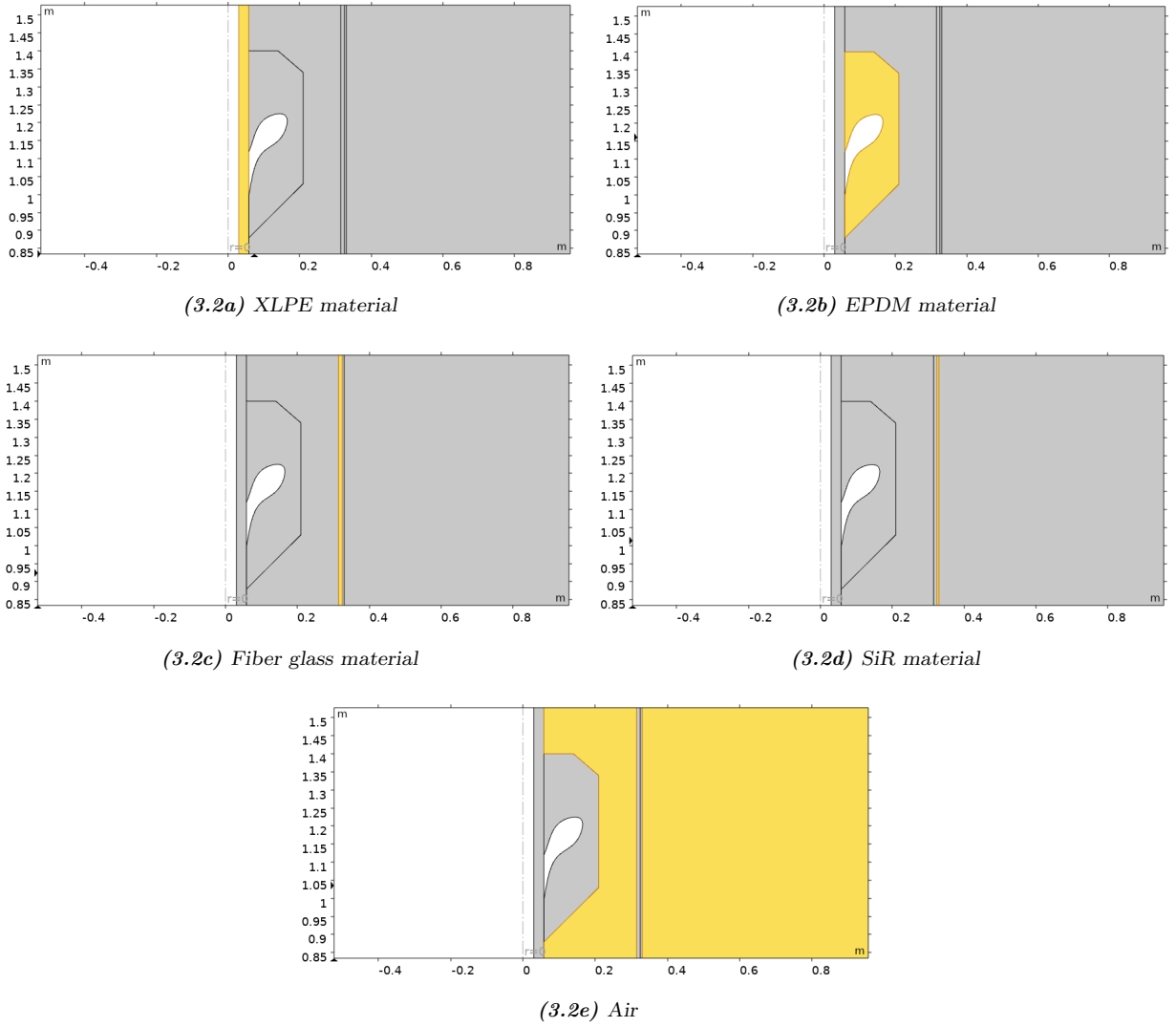


Figure 3.2: Zoomed-in view of the stress-relief cone

3.1.2 Materials and properties

Materials used and their properties are listed below in Table 3.1:

Material	Relative permittivity ε_r
EPDM	2.36
Fiber Glass	4.4
SiR	4.3
XLPE	2.3
Air	1

Table 3.1: Material properties

Boundary conditions: conductor inner surface assigned phase-to-ground potential; outer shield or enclosure grounded. Symmetry axis used for axisymmetric modelling. Interface continuity conditions applied between materials. A sufficiently fine mesh was defined in regions of high expected field (near the cone tip and screen termination).

3.2 Analysis Methodology: PDE and k -Values Criteria

In order to assess streamer inception within the gas region of the cable termination, the Partial Differential Method (PDE) method was implemented. This method combines a standard electrostatic field solution with a subsequent PDE evaluation of the ionization integral, allowing for an efficient and spatially resolved identification of streamer inception regions. [3]

1. Electrostatic field solution: The first step is to perform a standard electrostatic calculation to obtain the electric potential V and field $\mathbf{E} = -\nabla V$. For a geometry with fixed boundary conditions, the electrostatic field needs to be computed only once, and can later be reused for several streamer (K -value) calculations. In the simulation, the applied potential was normalized by setting $V = 1$, allowing results to be easily scaled to any other voltage level.
2. PDE setup in COMSOL: After the electrostatic study, a general PDE interface was added — specifically, the “**Stabilized Convection-Diffusion Equation**” interface in COMSOL Multiphysics. The stabilized version is necessary to obtain numerically stable FEM solutions, since pure convection problems tend to be unstable without artificial stabilization. This PDE interface uses the following stationary equation:

$$\nabla \cdot (-c \nabla K + \alpha \cdot K) + \beta \cdot \nabla K + \alpha K = f \quad (3.1)$$

Following the guidelines of the K -values methodology, all coefficients except β and f are set to zero. The convection term β is defined in the direction of the electric field, while f represents the local ionization coefficient. For spatial dimension 1, β_x is defined as:

$$\beta_x = \frac{k}{|\mathbf{E}|} \cdot E_x / (\varepsilon_0 \text{ normEeps}) \quad (3.2)$$

and similarly for other components, ensuring that the convection direction follows the local field vector. The parameter k is used to scale the solution. To avoid division by zero at field nulls, a very small constant “eps” is added in the denominator.

3. Boundary conditions: The boundary conditions are defined as follows:

- At the high-voltage electrode (cable conductor or semiconductive screen), $K = 0$;
- At the ground electrode or the outer boundary of the gas region, a Neumann (zero-gradient) boundary condition is used: $-\mathbf{n} \cdot \nabla K = 0$.

These conditions ensure that the integration of ionization starts at the electrode surface and that no artificial loss occurs at the outer boundary.

4. Computing and visualizing critical volumes: After solving the PDE, the computed field $K(x)$ represents the local cumulative ionization integral. Regions where $K(x) \geq K_c$ are identified as “critical volumes”, i.e. zones where streamer inception is possible.

The results are interpreted according to the following criteria:

- If the computed maximum $K_{\max} < K_c$ throughout the gas and along the surface, streamer inception is not expected under the applied voltage and pressure;
- If $K_{\max} \geq K_c$ within a contiguous region, this defines a potential streamer path — typically originating near the cone tip or the triple point where solid, gas, and electrode meet;
- The extent and volume of the $K \geq K_c$ region provide a measure of the risk severity;

Variation of pressure p and geometry parameters allows studying how the critical volume evolves — typically, increasing gas pressure reduces α_{eff} and hence the size of the critical region.

In summary, the PDE-based method allows for a direct computation of the streamer inception integral within the 2D/3D geometry. It provides a spatially resolved identification of regions at risk, enabling clear visualisation of “critical volumes” and offering a powerful complement to simple peak-field criteria.

3.3 Results and Comparative Analysis

In this section we present the comparative results obtained from the parametric study of three different stress-relief cone geometries (denoted Shape 0, Shape 1 and Shape 2), all made of EPDM, and evaluate their performance in terms of streamer-inception risk by analysing the critical volumes. A key feature of the procedure is the iterative parametric sweep: for each shape we first select a base gas pressure (e.g.,

1.0 bar), then vary the applied voltage until we observe the emergence of a region where the computed ionisation integral $k \geq K_c$. That voltage is denoted U_{incept} . If U_{incept} is below the reference voltage levels listed in Section 2, then the combination shape/pressure is deemed insufficient — the pressure is increased (to e.g., 2.0 bar) and the sweep repeated. This method ensures that for each shape we find the minimum pressure at which the termination geometry can sustain the required voltage levels without streamer inception risk.

3.3.1 Shape 0

- Tech-air pressure of 1 bar

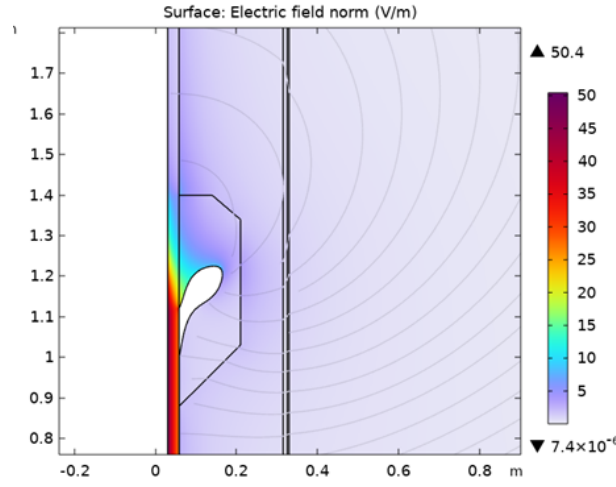


Figure 3.3: Electric field for the shape 0

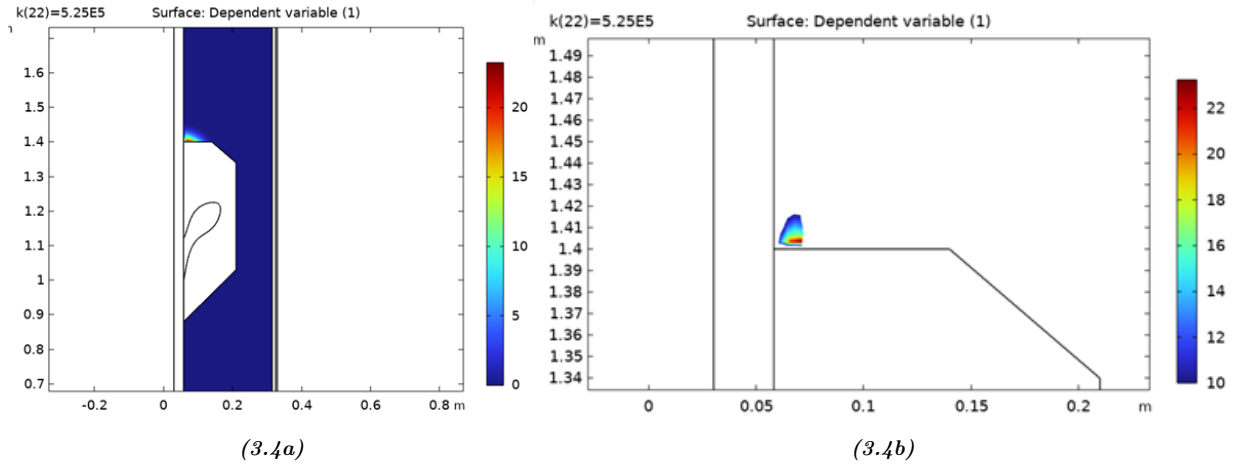


Figure 3.4: Overview of shape 0 at 1 bar

With tech-air at 1 bar, streamer inception occurs at c.a. 525 kV. The inception voltage is below the LI (1425 kV), SI (1175 kV) and TOV (630 kV) voltage levels specified in Section 2.

- Tech-air pressure of 2 bar

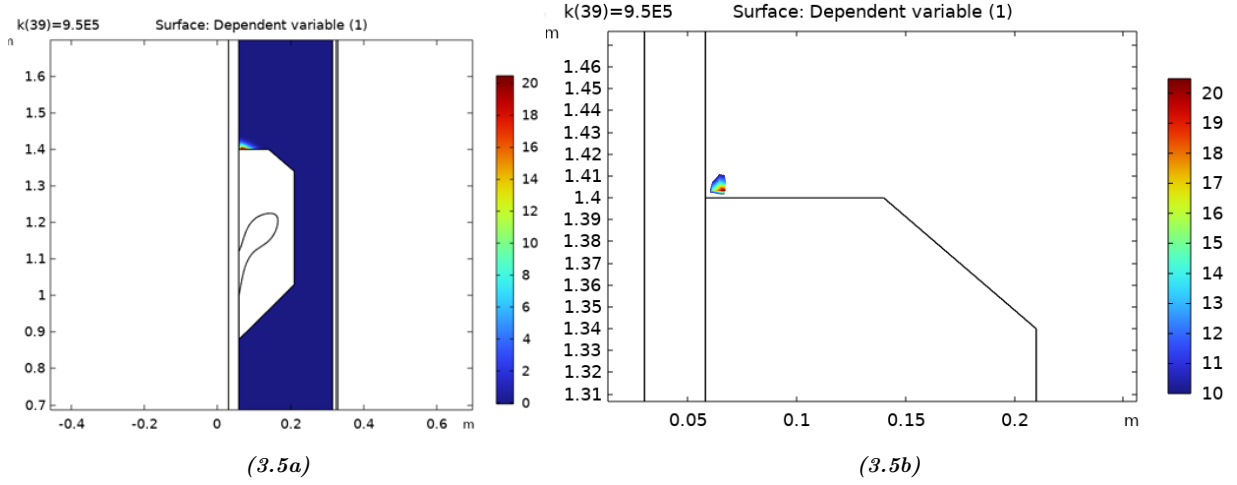


Figure 3.5: Overview of shape 0 at 2 bar

With tech-air at 2 bar, streamer inception occurs at c.a. 950 kV. The inception voltage is below the LI (1425 kV) and SI (1175 kV).

- Tech-air pressure of 3 bar

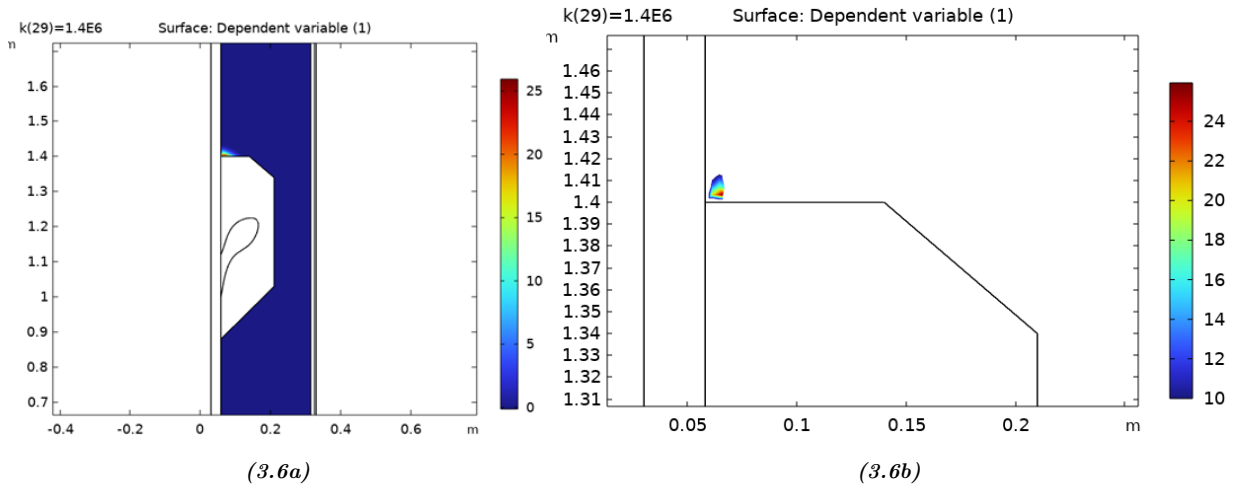


Figure 3.6: Overview of shape 0 at 3 bar

With tech-air at 3 bar, streamer inception occurs at c.a. 1400 kV. The inception voltage is only below the LI (1425 kV).

- Tech-air pressure of 4 bar

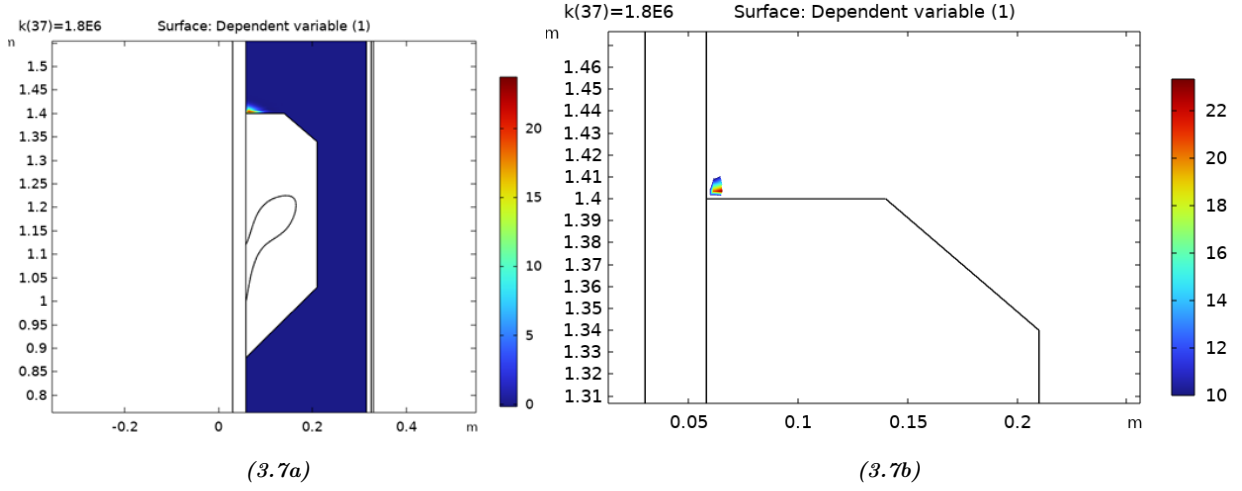


Figure 3.7: Overview of shape 0 at 4 bar

With tech-air at 1 bar, streamer inception occurs at c.a. 1800 kV. This configuration can satisfy all the voltage levels specified by IEC 60071-1 in Section 2.

3.3.2 Shape 1

- Tech-air pressure of 1 bar

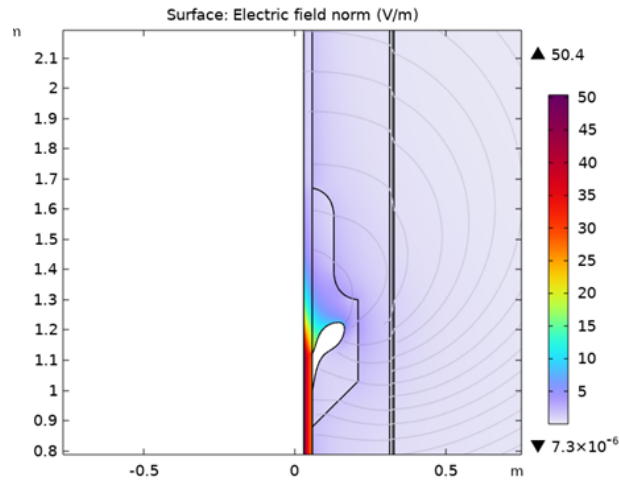


Figure 3.8: Electric field for the shape 1

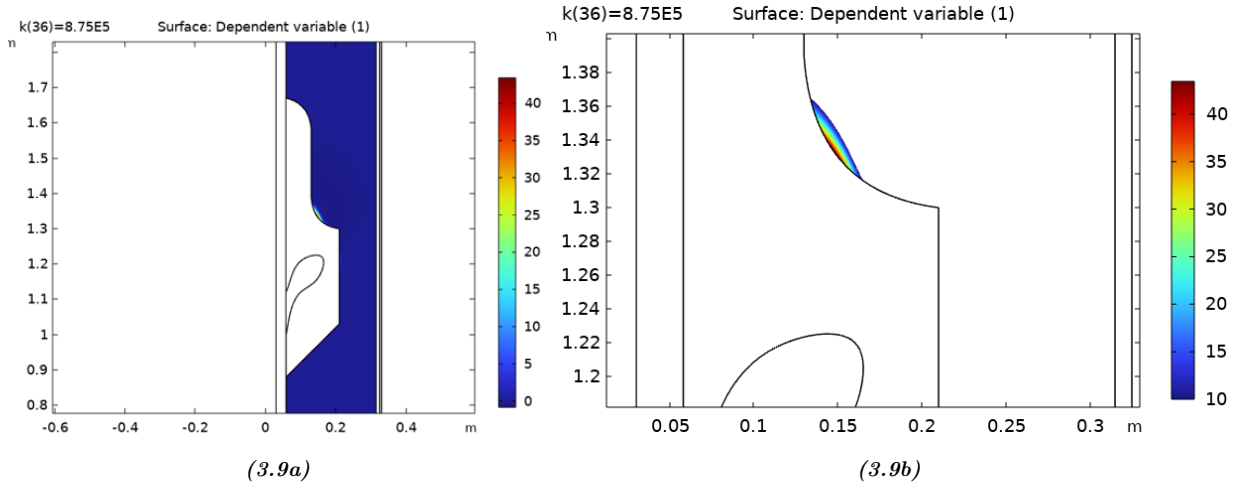


Figure 3.9: Overview of shape 1 at 1 bar

With tech-air at 1 bar, streamer inception occurs at c.a. 875 kV. The inception voltage is still below the LI (1425 kV) and SI (1175 kV).

- Tech-air pressure of 2 bar

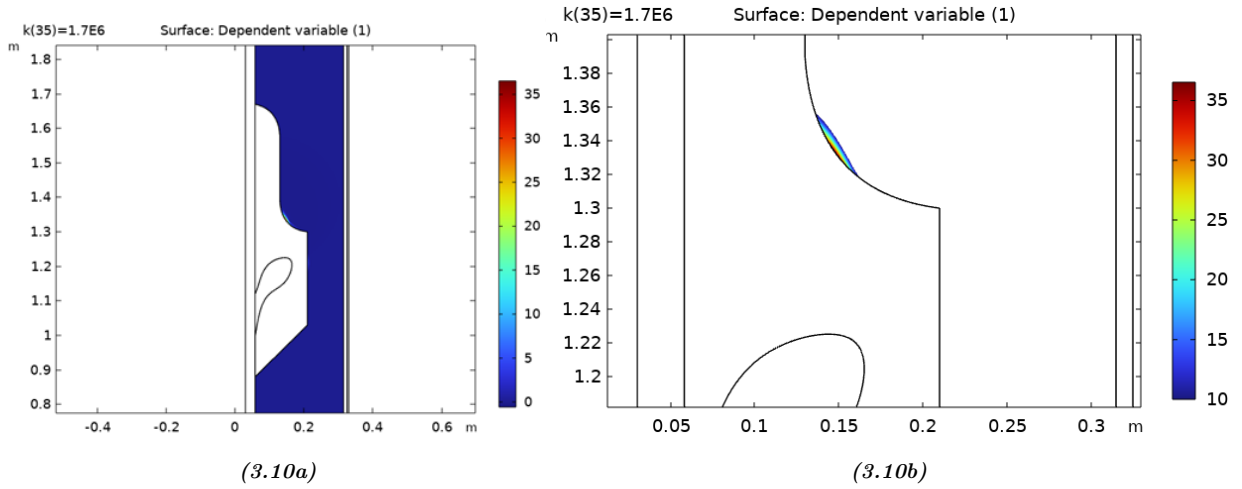


Figure 3.10: Overview of shape 1 at 2 bar

With tech-air at 2 bar, streamer inception occurs at c.a. 1700 kV. This configuration can satisfy all the voltage levels specified by IEC 60071-1 in Section 2.

3.3.3 Shape 2

- Tech-air pressure of 1 bar

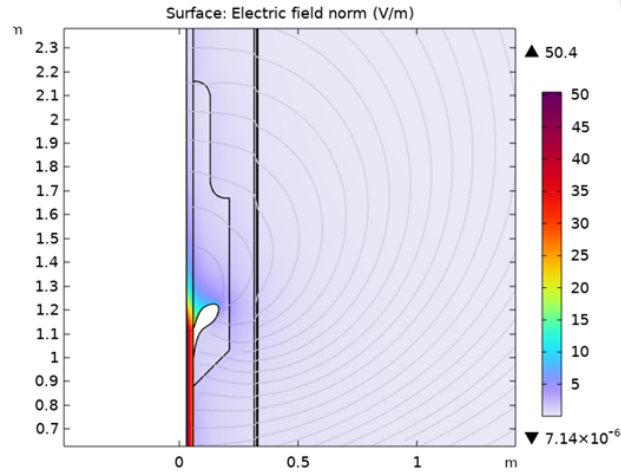


Figure 3.11: Electric field for the shape 2

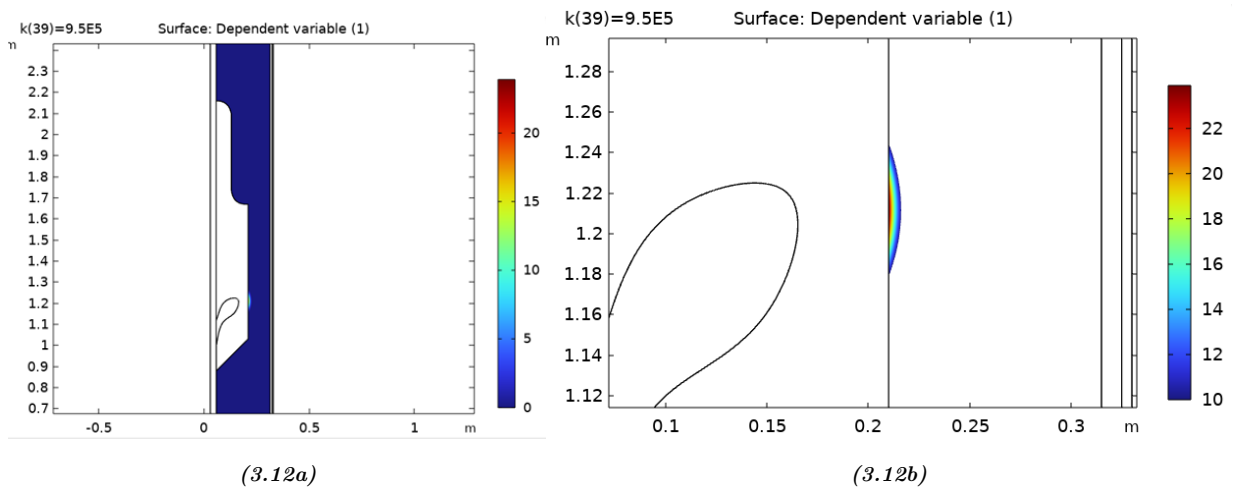


Figure 3.12: Overview of shape 2 at 1 bar

With tech-air at 1 bar, streamer inception occurs at c.a. 950 kV. The inception voltage is still below the LI (1425 kV) and SI (1175 kV).

- Tech-air pressure of 2 bar

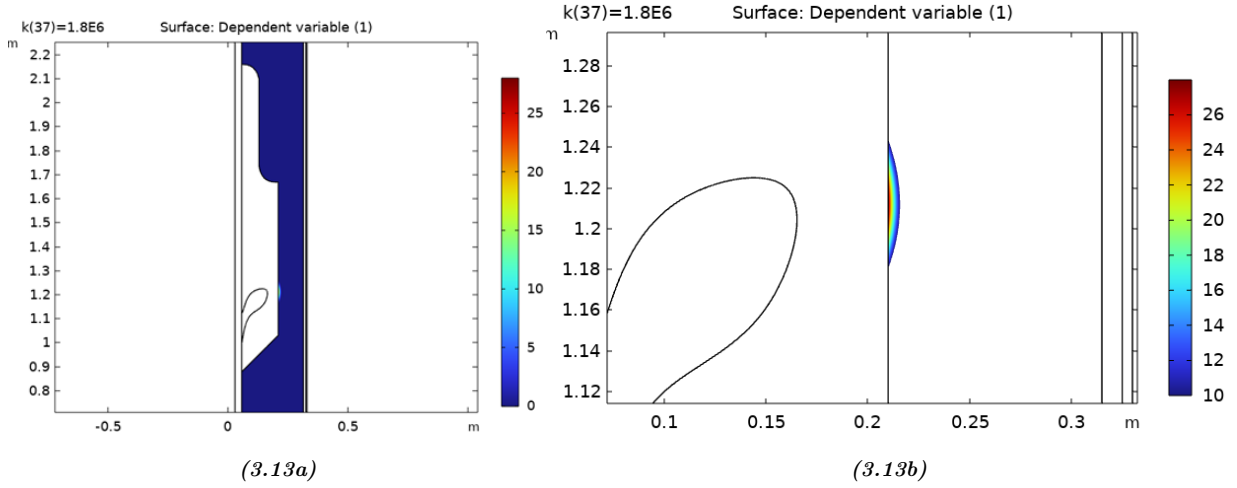


Figure 3.13: Overview of shape 2 at 2 bar

With tech-air at 2 bar, streamer inception occurs at c.a. 1800 kV. This configuration can satisfy all the voltage levels specified by IEC 60071-1 in Section 2.

3.3.4 Results summary

Shape	Pressure [bar]	U_{incept} [kV]	Comment
Shape 0	1.0	525	Inception at very low voltage $\rightarrow$ insufficient for any test levels
	2.0	950	Still below LI/SI thresholds $\rightarrow$ margin inadequate
	3.0	1400	Approaching but still just below LI $\rightarrow$ marginal design
	4.0	1800	Exceeds all reference voltages $\rightarrow$ acceptable only at high pressure
Shape 1	1.0	875	Better geometry but low pressure limits margin
	2.0	1700	Surpasses all voltage levels $\rightarrow$ viable configuration
Shape 2	1.0	950	Best shape but pressure too low for safe margin
	2.0	1800	Safely exceeds all test levels at moderate pressure

Table 3.2: Inception voltages U_{incept} for each shape and pressure level

From Table 3.2 we observe the following:

- Shape 0 at 1.0 and 2.0 bar clearly fails to meet the voltage levels set out in Section 2, indicating that under those conditions streamer inception could occur under normal or test voltages.
- Shape 1 at 2.0 bar provides an inception voltage of 1700 kV, which exceeds all specified voltage levels, making it a strong candidate for reliable performance.
- Shape 2 at 2.0 bar offers the best margin, with inception voltage (1800 kV) well beyond test conditions. Therefore it is the best solution if installation and cost allow.

3.3.5 Design recommendation

Based on the comparative analysis, the following recommendations are made: - The stress-relief cone geometry should favour Shape 1 or Shape 2. - A tech-air pressure of at least 2.0 bar is required to achieve inception voltages that exceed the reference test levels. - If mechanical/installation constraints are restrictive, Shape 1 at 2.0 bar is chosen as the balanced solution; if maximum margin is required and space allows, Shape 2 at 2.0 bar (or higher) is preferred. - Shape 0 only becomes acceptable at 4.0 bar, which may impose impractical pressure or installation demands.

With these results the modelling and analysis phase supports a clear decision path toward the final design.

4 Conclusion

In this project we have investigated the electric-field grading and streamer-inception behaviour of a 500 kV cable termination employing a compressed tech-air insulation medium. The main objectives were to understand the underlying theory of high-voltage terminations and insulation media, to develop a finite-element model in COMSOL representing the termination geometry (including an EPDM stress-relief cone), to apply a PDE-based method via the k -value (ionization integral) criterion to assess streamer inception risk, and finally to perform a parametric study of three cone geometries under varying gas pressures in order to derive a design recommendation.

Such analyses are critically important because, for ultra-high voltage systems, the reliability of cable terminations depends heavily on effective electric-field control, prevention of partial discharges and streamers, and insulation coordination in accordance with standards such as IEC 60071-1.

Throughout the report we first presented the background theory (construction of cable terminations, insulation media, stress-relief cones, streamer and surface-breakdown theory), then described the model construction and analysis methodology (COMSOL geometry and materials, PDE implementation, k -value criteria), and finally provided results from a comparative parametric study of three cone shapes and tech-air pressures.

The parametric sweep carried out in this study provided clear insights into the behaviour of the different geometrical configurations of the stress-relief cone under varying tech-air pressures. Shape 0 exhibited inception voltages significantly below the required test levels at lower pressures, and only reached acceptable values at elevated pressure (making it less practical under standard conditions). Shape 1, with a more open geometry, achieved inception voltages beyond all reference levels at a moderate pressure (2 bar), and thus appears as the most practical and balanced solution. Shape 2, being the most open geometry considered, offered the highest margin by reaching inception voltages well beyond test levels already at moderate pressure.

Accordingly, the design recommendation is to adopt a stress-relief cone geometry corresponding to Shape 1 or Shape 2, together with a tech-air pressure of at least 2.0 bar. If maximum margin is required and available space allows, Shape 2 at 2.0 bar (or higher) is preferred. The simpler geometry of Shape 0 becomes acceptable only under much higher gas pressure (4.0 bar), which may impose impractical demands for pressure containment or installation.

In summary, the modelling and analysis procedure provides a clear decision path toward the final termination design: by selecting an appropriate combination of cone geometry and gas pressure, the termination can safely withstand nominal, impulse, switching and temporary over-voltage stresses without the risk of streamer inception.

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